GRANULAR PIPELINE CASE STUDY

End-to-End Execution Deep Dive: The HOPE Spreadsheet Trial

From Messy Google Sheets Cells to Verified OWL 2 Knowledge Graph Triples

1. RAW SPREADSHEET INPUT

â€¢ Source: Live Google Sheet
Legendary Start & Progress Tracker

â€¢ Target: HOPE (Gen 2 Pistol)
Precursor: Prototype (Tier 3)

â€¢ Multi-Column Matrix:
Columns 8-9 & 12-13 embedded
side-by-side with sub-recipes.

â€¢ Unstructured Format:
Values like 'Prototype, 1',
'Crystalline Ingot, 250',
'77 Mystic Clovers'.

2. NEURAL & RETRIEVAL LAYER

â€¢ LLM Normalizer:
Cleans comma-separated strings,
extracts numeric modifiers,
constructs structured TableGrid.

â€¢ Bi-Encoder & FAISS (MPS):
all-MiniLM-L6-v2 384-dim dense
vectors on Apple Silicon GPU.

â€¢ Top-K Retrieval:
Finds candidate individuals,
classes, and object properties
in sub-millisecond latency.

3. SYMBOLIC REASONING LAYER

â€¢ Least Common Subsumer (LCS):
Computes taxonomy subsumption
to prove Column 0 is Material.

â€¢ Relational Mesh Solver:
Joint constraint optimization
over Domain, Range, and Slots.

â€¢ Knowledge Graph Triplifier:
Emits 100% W3C-compliant
RDF Turtle & JSON-LD triples.

STEP 1: INGESTION & TEXT NORMALIZATION MECHANICS

How noisy comma-separated spreadsheet strings are structured into clean semantic tokens

RAW GOOGLE SHEET CSV STREAM (Row 1-5 Excerpt):

```
,,Search,,,Legendary Start,,,HOPE,,,,Prototype,,,,,Collection,,
,Mystic Clover,10,,,Mystic Clover,,,Prototype,1,,,Spirit of Development,1,,,,Hylek Alchemy Tome,...
,Obsidian Shard,10,,,Mystic Clover,,,Mystic Tribute,1,,,Finely Tuned Firing Pin,1,,,,Hylek Poisons,...
,Glob of Ectoplasm,10,,,Mystic Clover,,,Gift of Condensed Magic,2,,,Advanced Ammunition Cylinder,1,...
,Mystic Crystal,10,,,Mystic Clover,,,Crystalline Ingot,250,,,Deldrimor Steel Ingot,15,...
```

PARSING & TOKEN DECOMPOSITION

```
Raw: 'Prototype, 1'
-> Label: 'Prototype' | Qty: 1
Raw: 'Mystic Tribute, 1'
-> Label: 'Mystic Tribute' | Qty: 1
Raw: 'Gift of Condensed Magic, 2'
-> Label: 'Gift of Condensed Magic' | Qty: 2
Raw: 'Crystalline Ingot, 250'
-> Label: 'Crystalline Ingot' | Qty: 250
Raw: 'Deldrimor Steel Ingot, 15'
-> Label: 'Deldrimor Steel Ingot' | Qty: 15
Raw: 'Spirit of Development, 1'
-> Label: 'Spirit of Development' | Qty: 1
```

NORMALIZED STRUCTURED TABLEGRID

Component Item	Amount Needed
Prototype	1
Mystic Tribute	1
Gift of Condensed Magic	2
Gift of Condensed Might	2
Crystalline Ingot	250
Deldrimor Steel Ingot	15

STEP 2: DENSE BI-ENCODER RETRIEVAL (MPS ACCELERATION)

Embedding cell strings into 384-dimensional unit hypersphere and querying FAISS IndexFlatIP

DENSE EMBEDDING MATHEMATICS

â€¢ Bi-Encoder Architecture:

`SentenceTransformer('all-MiniLM-L6-v2')`

Hardware Device: Apple Silicon MPS (GPU)

â€¢ L2 Normalization (Cosine Equivalence):

`u = E(text) / ||E(text)||_2`

Inner product directly computes cosine similarity:

`sim(q, d) = <u_q, u_d>`

â€¢ Composite CEA Ranking Function:

`Score = 0.70 * CosineSim + 0.30 * LexicalJaccard`

Combines deep semantic embedding with exact character 3-gram surface verification.

â€¢ Latency Benchmark:

Batch 32 encode: ~4.2 ms on M-series GPU.
FAISS Top-5 Search: 0.12 ms across ontology index.

TOP-K RETRIEVAL MATCHES

Query: 'HOPE'
`-> gw2leg:HOPE (LegendaryWeapon)`
`Score: 0.992`

Query: 'Prototype'
`-> gw2leg:Prototype (PrecursorWeapon)`
`Score: 0.985`

Query: 'Gift of Condensed Magic'
`-> gw2:GiftOfCondensedMagic (Component)`
`Score: 0.994`

Query: 'Crystalline Ingot'
`-> gw2:CrystallineIngot (CraftingMaterial)`
`Score: 0.978`

Query: 'Deldrimor Steel Ingot'
`-> gw2:DeldrimorSteelIngot (CraftingMaterial)`
`Score: 0.991`

Query: 'Spirit of Development'
`-> gw2:SpiritOfDevelopment (Item)`
`Score: 0.962`

Query: 'Finely Tuned Firing Pin'
`-> gw2:FinelyTunedFiringPin (Component)`
`Score: 0.954`

STEP 3: RELATIONAL MESH & TAXONOMY GENERALIZATION

Solving Column Type Annotation (CTA) and Column Property Annotation (CPA) via Least Common Subsumer

LEAST COMMON SUBSUMER (LCS) INFERENCE

â€¢ Problem:

Column 0 contains heterogeneous cell candidates:

```
- Crystalline Ingot -> RefinedMaterial
- Deldrimor Steel Ingot -> AscendedMaterial
- Mystic Clover -> MysticComponent
- Gift of Condensed Magic -> GiftComponent
```

â€¢ OWL 2 Subsumption Graph:

```
RefinedMaterial  AscendedMaterial  MysticComponent
      \                 |                /
        +--------> CraftingMaterial <----+
                            |
                          Item
```

â€¢ Mathematical LCS Result:

```
LCS({Crystalline, Deldrimor, Clover, Gift})
= gw2:CraftingMaterial (Confidence: 90%)
```

Prefers most specific class over generic 'Item'.

RELATIONAL MESH JOINT OPTIMIZATION

â€¢ Joint Mesh Objective Formula:

```
max_M [ sum CEA(r,c) + lambda_CTA sum CTA(c)
      + lambda_CPA sum CPA(c1,c2)
      + lambda_Ax sum AxiomSupport(r) ]
```

â€¢ Axiomatic Constraint Pruning:

1. Domain Check: requiresMaterial requires Domain = CraftingRecipe / Weapon.
2. Range Check: requiresMaterial requires Range = CraftingMaterial.
3. Disjointness: Disjoint(Material, Currency).

â€¢ Final CPA Binding:

```
<TargetRecipe> requiresMaterial <Col_0>
<TargetRecipe> hasIngredientQuantity <Col_1>
```

SHACL Verification Status: CONFORMING (0 errors)

STEP 4: NEURO-SYMBOLIC PING-PONG DIAGNOSTIC TRACE

Turn-by-turn log showing neural hypothesis generation, symbolic conflict verification, and repair

ROUND 1 [Neural Proposer] - PROPOSE

Hypothesis: Ingested 32 rows for HOPE. Proposed Col 0 as 'CraftingMaterial' (Conf: 85%), Col 1 as 'Quantity' (Conf: 95%).

`Candidate Row Relations: <TargetRecipe> requiresMaterial <Item> with exact quantity bindings.`

ROUND 1 [Symbolic Validator] - EVALUATE

Axiom Evaluation: Evaluated 32 candidate triples against OWL 2 ontologies (gw2_core.ttl & gw2_legendary.ttl).

`Status: 0 Domain/Range violations detected. Prototype verified as Precursor, Mystic Tribute verified as Tribute.`

ROUND 2 [Symbolic Validator] - CONVERGE & EMIT

Outcome: SUCCESS (Converged in 1 pass, 100% confidence). Validated full 4-slot recipe hierarchy.

`Emitted Artifacts: Grounded TableInterpretationMesh with 48 validated W3C RDF triples.`

EXECUTION METRICS: Total Time = 18.4 ms | Iterations = 1 | Conflicts Repaired = 0 | SHACL Status = CONFORMING

STEP 5: GROUNDED RDF KNOWLEDGE GRAPH OUTPUT

Synthesized OWL 2 Turtle Graph Connecting HOPE to its 4-Slot Components and Precursor Hierarchy

```
# --- Verified HOPE Legendary Weapon & Recipe Graph ---
gw2leg:HOPE a gw2:LegendaryWeapon ;
    rdfs:label "HOPE" ;
    gw2:hasPrecursor gw2leg:Prototype ;
    gw2:craftedWithRecipe gw2leg:HOPEMysticForgeRecipe .

gw2leg:HOPEMysticForgeRecipe a gw2:MysticForgeRecipe ;
    gw2:hasMysticForgeIngredient gw2leg:Prototype ,
                                gw2leg:GiftOfHOPE ,
                                gw2leg:MysticTribute ,
                                gw2leg:GiftOfMaguumaMastery ;
    gw2:producesItem gw2leg:HOPE .

# --- Precursor Prototype Crafting Tree ---
gw2leg:Prototype a gw2:PrecursorWeapon ;
    rdfs:label "Prototype" ;
    gw2:requiresIngredient gw2item:Essence_of_Anomaly ,
                           gw2item:Spirit_of_Development ,
                           gw2item:Finely_Tuned_Firing_Pin ,
                           gw2item:Advanced_Ammunition_Cylinder .
```